

Graph Convolutional Network and Transformer for Molecular Representation Learning and Generation

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Abstract

This work jointly learns representations of molecular graphs and SELFIES sequences on the MoleculeNet Lipophilicity dataset. A three-layer, bond-order-weighted residual graph convolutional network is aligned with a four-layer causal Transformer using symmetric contrastive and positive-pair objectives while retaining next-token language modeling. The resulting Transformer is sampled autoregressively at three temperatures, and every output is subjected to staged RDKit validation, canonical novelty measurement, Morgan fingerprint comparison, and graph-sequence representation analysis. This is the assignment-compliant execution with 15 language-model warm-up epochs and 300 generation attempts per temperature; joint training used validation monitoring and executed 40 epoch(s) (cap 40, patience 7).

1 Introduction

Molecular graphs and line notations expose complementary inductive biases. Graph convolution directly aggregates local atomic neighborhoods, whereas a causal sequence model learns a distribution over syntactically ordered SELFIES symbols. The experiment asks whether a shared contrastive space can associate these views and whether the language model can generate structures that survive a conservative cheminformatics screen. All stochastic components use seed 42; the fixed data split and vocabulary are saved with the results.

2 Dataset Preparation

PyTorch Geometric’s `MoleculeNet` loader supplied the Lipophilicity dataset. The installed PyG release indexed it by the explicit alias `Lipo`; the loader first attempted `Lipophilicity` and recorded the fallback. Each original SMILES was parsed without sanitization, sanitized explicitly, and converted to canonical isomeric SMILES. Canonical strings were deterministic deduplication keys. The first representative retained its original lipophilicity target. SELFIES conversion used the official encoder and splitter. The 100-symbol limit was applied to raw SELFIES symbols, excluding `<bos>` and `<eos>`.

Table 1: Measured dataset statistics.

Statistic	Measured value
Original molecules	4200
Removed molecules	6
Canonical duplicates	0
Usable molecules	4194
Vocabulary size	52
Mean atom count	26.9664
Mean undirected bond count	29.4242
SELFIES length: min / mean / median / max	9 / 44.39 / 45.0 / 94

Table 2: Cleaning and filtering removals.

Removal reason	Count
invalid_smiles	0
sanitization_failure	0
canonicalization_failure	0
duplicate_canonical	0
selfies_conversion_failure	0
selfies_too_long	6

Table 3: Fixed deterministic split sizes.

Split	Molecules
Training	3355
Validation	419
Test	420

A seed-42 permutation produced approximate 80/10/10 splits after all cleaning. Canonical sets were asserted disjoint. Vocabulary counts used only training SELFIES and included exactly `<pad>`, `<bos>`, `<eos>`, and `<unk>`. Validation contained 3 unknown symbol occurrence(s) (0.0161%), and test contained 0 (0.0000%). Figure 1 shows the measured sequence-length distribution.

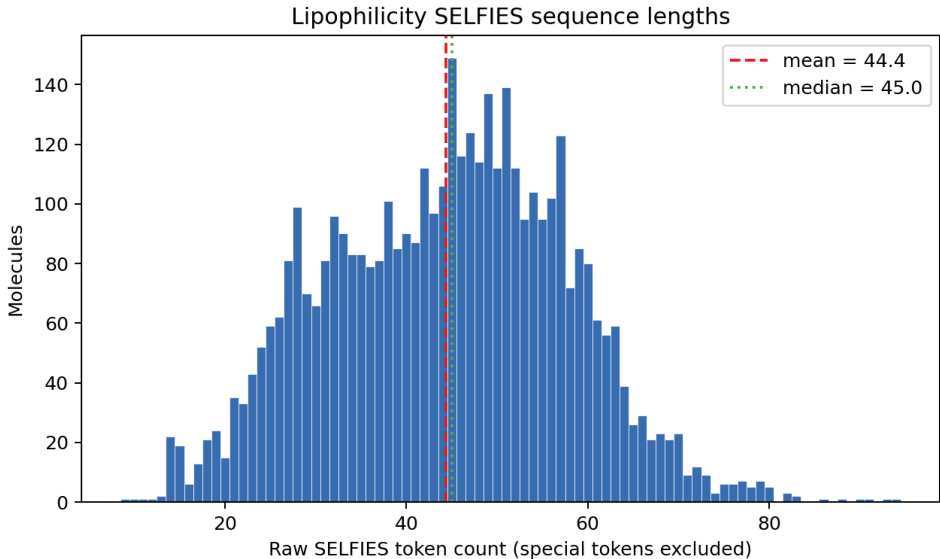


Figure 1: Raw SELFIES sequence lengths after cleaning.

3 Graph Representation and GCN

Canonical molecules were reconstructed with PyG’s official SMILES converter. Its nine categorical atom fields are atomic number, chirality, degree, formal charge, hydrogen count, radical-electron count, hybridization, aromaticity, and ring membership. Each field has a separate embedding table with an additional unknown category; equal-width field embeddings are added elementwise. This preserves categorical identities without concatenation.

For each undirected RDKit bond, both directions were stored with scalar weight 1.0 (single), 2.0 (double), 3.0 (triple), or 1.5 (aromatic). Exactly three `GCNConv` blocks apply weighted normalized convolution, layer normalization, GELU, dropout, and residual addition. Global mean pooling is necessary because molecules have variable numbers of atoms, whereas retrieval and contrastive learning require one fixed-dimensional vector per molecule. A two-layer graph projection head maps the pooled vector to 128 dimensions and applies L_2 normalization.

A scalar bond weight is only an approximation: one number cannot jointly encode bond type, stereochemistry, conjugation, aromatic and ring context, directionality, and other chemical attributes. It is not equivalent to a complete edge-feature message-passing model.

4 SELFIES Representation and Transformer

The causal Transformer has learned token and position embeddings, four encoder layers, six self-attention heads, hidden dimension 192, feedforward dimension 512, and dropout 0.1. Its 102 learned positions support `<bos>`, 100 raw symbols, and `<eos>`. An upper-triangular mask hides future positions and a key padding mask excludes padded keys. Padding targets are ignored by cross-entropy. Causal masking prevents next-token information leakage and makes teacher-forced training consistent with autoregressive inference.

Given $\langle \text{bos} \rangle, t_1, \dots, t_m, \langle \text{eos} \rangle$, inputs predict the one-position-shifted targets. Teacher forcing exposes the correct prefix during training; generation later repeats the learned conditional next-token distribution using its own sampled prefix. The hidden state at the actual EOS position is the

sequence-level molecular representation, followed by a distinct two-layer normalized projection head. If EOS is absent during incomplete inference, the final generated-token state is used.

5 Joint Representation Learning

For non-padding targets y , token cross-entropy is

$$\mathcal{L}_{\text{token}} = -\frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \log p_{\theta}(y_{it} \mid y_{i,<t}).$$

Let $G, S \in \mathbb{R}^{B \times D}$ be normalized graph and sequence projections and $Z = GS^{\top} / \tau$, with $\tau = 0.07$. For diagonal labels $a_i = i$,

$$\begin{aligned} \mathcal{L}_{g \rightarrow s} &= \text{CE}(Z, a), & \mathcal{L}_{s \rightarrow g} &= \text{CE}(Z^{\top}, a), \\ \mathcal{L}_{\text{contrastive}} &= \frac{1}{2}(\mathcal{L}_{g \rightarrow s} + \mathcal{L}_{s \rightarrow g}), & \mathcal{L}_{\text{pair}} &= \frac{1}{B} \sum_i (1 - G_i^{\top} S_i). \end{aligned}$$

The optimized joint objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{token}} + \lambda \mathcal{L}_{\text{contrastive}} + \gamma \mathcal{L}_{\text{pair}}, \quad \lambda = 1.0, \quad \gamma = 0.25.$$

A batch of B pairs supplies one positive and approximately $B - 1$ in-batch negatives per query. Larger batches therefore improve negative coverage, but cost activation and similarity-matrix memory.

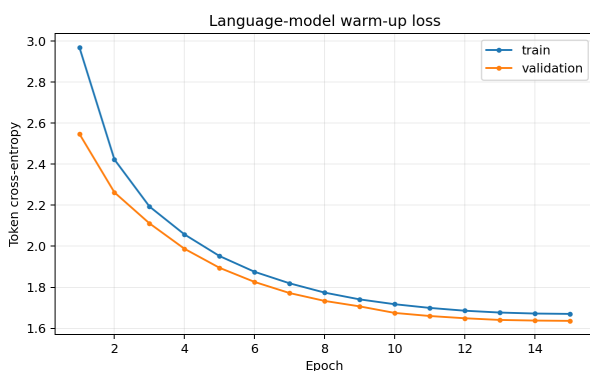
6 Training Procedure

Language-model warm-up first trained only the Transformer embeddings, blocks, normalization, and tied vocabulary head. This stage lets the model acquire SELFIES syntax and local patterns before multimodal alignment. Joint training then updated the GCN, Transformer, both projection heads, and language-model head. Both stages used AdamW, cosine decay, gradient norm clipping at 1.0, and CUDA automatic mixed precision when CUDA was available. The best warm-up checkpoint minimized validation token loss; the best joint checkpoint minimized validation total loss with patience 7. Table 4 gives all exact settings.

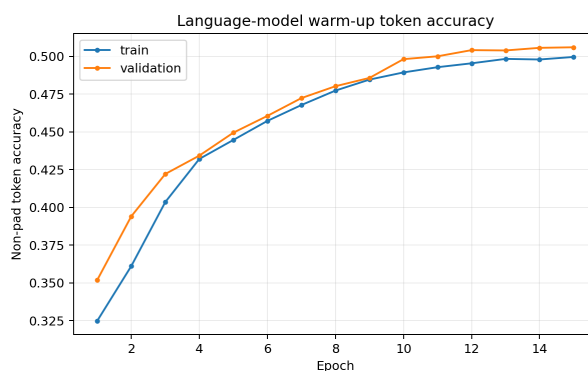
Table 4: Model and optimization settings.

Hyperparameter	Value
Batch size	256
GCN hidden dimension / layers	192 / 3
GCN dropout	0.1000
Transformer dimension / heads / layers	192 / 6 / 4
Transformer feedforward dimension	512
Transformer dropout	0.1000
Projection dimension	128
Contrastive temperature	0.0700
Lambda / gamma	1.0 / 0.25
Warm-up epochs: default / executed	15 / 15
Joint epochs: maximum / executed	40 / 40
Warm-up / joint learning rate	0.0003 / 0.0002
Generation attempts per temperature: default / executed	300 / 300
Top-k / nucleus p	20 / 0.95

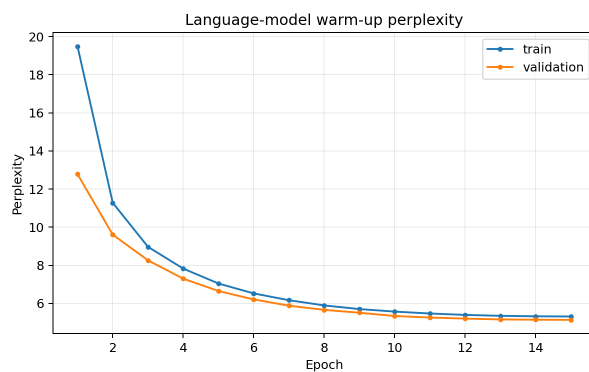
The best measured warm-up epoch was 15; the best joint epoch was 38, with validation total loss 2.5053, token loss 1.6158, token accuracy 0.5153, and perplexity 5.032. Figures 2 and 3 show the measured curves.



(a) Token loss.



(b) Non-padding accuracy.



(c) Perplexity.

Figure 2: Language-model warm-up measurements.

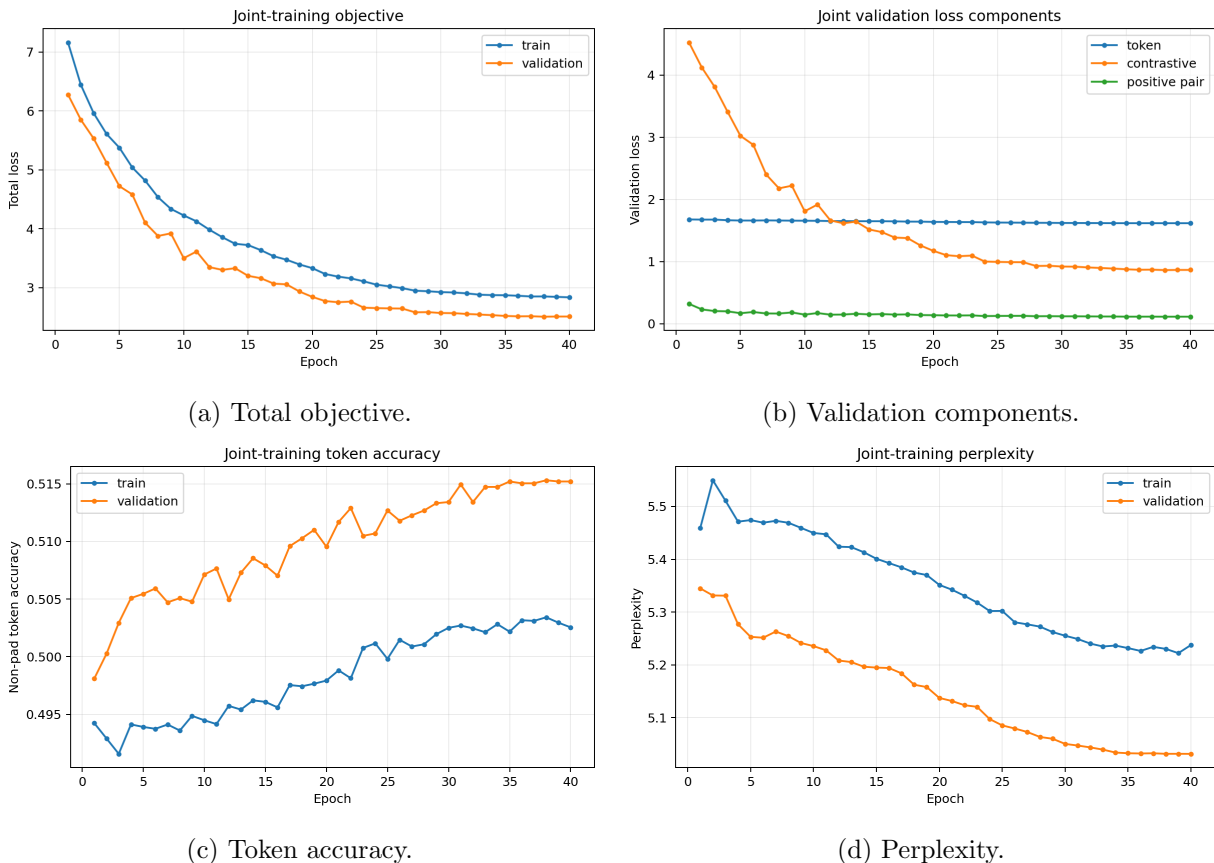


Figure 3: Joint-training measurements.

7 Cross-Modal Retrieval Results

The best joint checkpoint encoded all 420 test molecules once. Table 5 reports both retrieval directions. For N test pairs, random top-1 is $1/N$, and random top-5 is $\min(5/N, 1)$.

Table 5: Bidirectional test retrieval and similarity results.

Metric	Value
Graph to sequence top-1	0.8190
Graph to sequence top-5	0.9905
Sequence to graph top-1	0.7905
Sequence to graph top-5	0.9810
Random top-1	0.0024
Random top-5	0.0119
Matching cosine mean	0.8922
Matching cosine standard deviation	0.0510
Nonmatching cosine mean	0.0129
Nonmatching cosine standard deviation	0.3359

The measured matching-minus-nonmatching cosine gap was 0.8793. Both top-1 directions exceeded the finite-test-set random rate. Figure 4 visualizes a fixed aligned subset; a strong model should produce a clear diagonal.

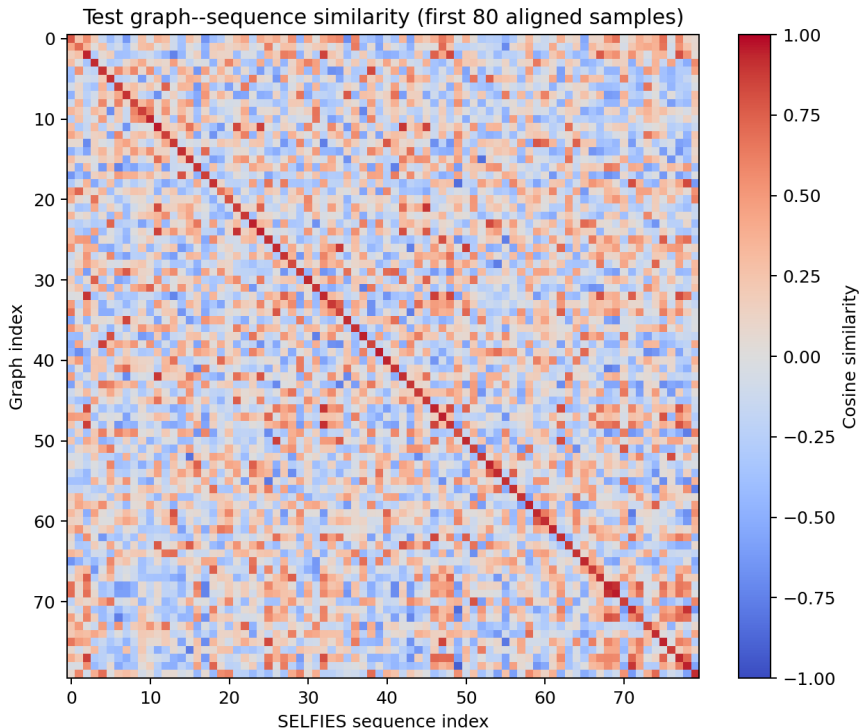


Figure 4: Test graph-SELFIES cosine-similarity matrix.

8 Molecular Generation

Each attempt began with only `<bos>` and stopped at `<eos>` or 100 raw symbols. Sampling banned `<pad>`, `<bos>`, and `<unk>` throughout and banned EOS for the first five raw symbols. Logits were temperature-scaled at 0.8, 1.0, and 1.2, restricted to top-20, and nucleus-filtered at $p = 0.95$. Distinct seeded streams preserved reproducibility.

9 Generated-Molecule Validation

Validation applied, in order: (1) SELFIES decoding; (2) RDKit parsing with sanitization disabled; (3) explicit RDKit sanitization; (4) a one-component requirement; (5) training-element and no-dummy-atom checks; (6) zero radical electrons; (7) the training 1st–99th percentile heavy-atom bound; (8) the separately evaluated training 1st–99th percentile molecular-weight bound; (9) the training maximum absolute molecular formal charge; and (10) canonical isomeric SMILES conversion. RDKit-valid counts require stages 2–3; accepted counts require every stage. The measured heavy-atom interval was [11.000, 43.000], molecular-weight interval [148.189, 616.751] Da, maximum absolute formal charge 1, and allowed elements B, Br, C, Cl, F, I, N, O, P, S, Se, Si.

Table 6: Generation metrics with explicit attempt counts.

T	N	Valid	Valid rate	Accepted	Accept rate	Unique	Unique rate	Novel	Novel rate	Struct.	Struct. rate	Mean QED	Mean length
0.8000	300	300	1.0000	273	0.9100	273	1.0000	273	1.0000	273	1.0000	0.5187	40.8000
1.0000	300	300	1.0000	250	0.8333	250	1.0000	250	1.0000	250	1.0000	0.4801	43.8233
1.2000	300	300	1.0000	183	0.6100	183	1.0000	183	1.0000	183	1.0000	0.4203	43.7800

Table 7: First-failure rejection counts by validation stage.

First-failure stage	T=0.8	T=1.0	T=1.2
accepted	273	250	183
canonicalization	0	0	0
connected_components	0	0	0
elements_and_dummy_atoms	0	0	0
formal_charge	0	0	0
heavy_atom_range	25	38	67
molecular_weight_range	2	3	1
radical_electrons	0	9	49
rdkit_parse	0	0	0
rdkit_sanitise	0	0	0
selfies_decode	0	0	0

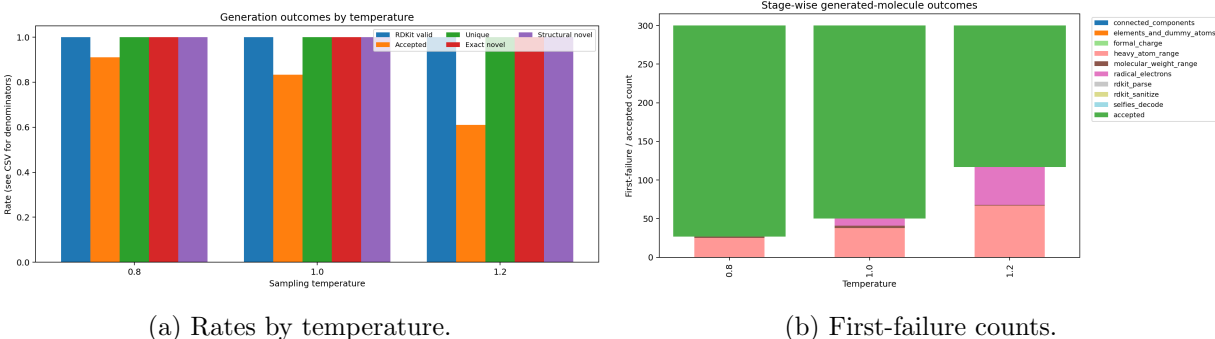


Figure 5: Measured generation and validation outcomes.

Temperature 0.8 had the highest measured RDKit-valid rate (1.000); temperature 0.8 had the largest measured structurally novel count (273). Lower temperature often makes sampling more conservative and higher temperature can increase both diversity and errors, but interpretation here is restricted to these measured counts and does not assume that generic trend.

10 Novelty and Fingerprint Analysis

Canonicalization is essential because multiple SMILES strings can denote the same molecular structure. Uniqueness therefore divides distinct accepted canonical structures by accepted samples,

and exact novelty tests unique accepted structures against the cleaned training canonical set. Exact novelty alone does not prevent near-memorization: a new canonical structure may differ by a tiny substitution while remaining extremely similar to a training molecule. Radius-two, 2,048-bit Morgan fingerprints quantify this relationship. An exact-novel structure is called structurally novel only when its maximum training-set Tanimoto similarity is below 0.85.

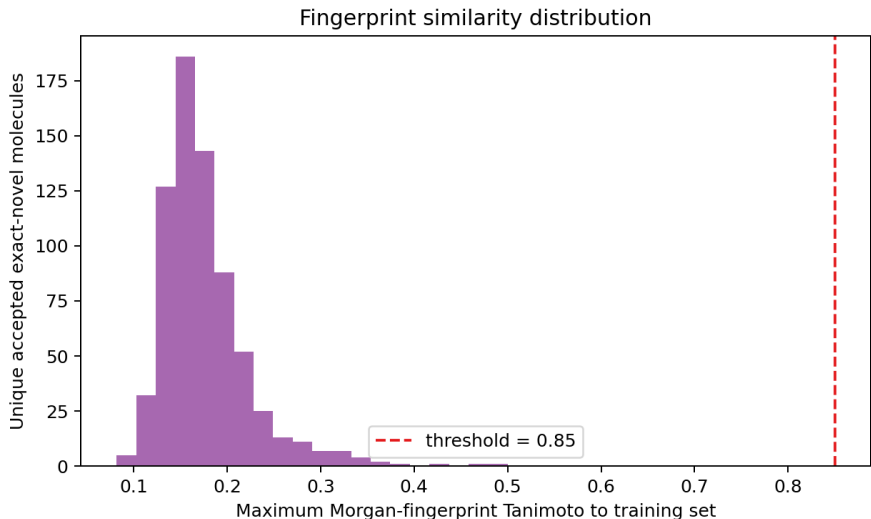


Figure 6: Maximum training-set Morgan-fingerprint similarity for unique accepted exact-novel structures.

Mean QED in Table 6 is descriptive only. QED does not establish usefulness, safety, synthesis feasibility, or biological activity.

11 Generated-Molecule Representation Analysis

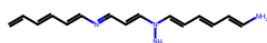
Every accepted exact-novel molecule was converted with the same canonical graph encoder and represented by its generated SELFIES. Graph-sequence compatibility is their projected cosine. The combined representation was $\text{normalize}(g + s)$. Each generated vector was compared with precomputed combined training vectors. Table 8 reports up to ten deterministic examples; if fewer were produced, it reports all available rather than inventing rows.

Table 8: Up to ten accepted exact-novel generated molecules and representation-space neighbors.

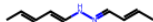
Generated	T	Graph-seq.	Nearest training	Embedding sim.	Max Tanimoto	QED
<chem>CC(=O)Nc1cc(=O)c2c3ccc(cc[nH]ccc1=2)</chem>	1.0000	0.9492	<chem>CNC(=O)c1ccc2c(=O)cc(-c3ccccc3)[nH]c</chem>	0.9657	0.2800	0.7165
<chem>CC(N)CCCCc1c2ccccccc1-2</chem>	0.8000	0.9284	<chem>CC(C)COCC(CN(Cc1cccc1)c1cccc1)N1CC</chem>	0.9693	0.3158	0.7713
<chem>Nc1cc2ccc1[nH]ccc1c3ccc(cc3)c21</chem>	0.8000	0.9198	<chem>c1ccc(-c2cn3cccc3n2)cc1</chem>	0.9542	0.3333	0.4410
<chem>CC1CC2C=C3C4=Cc5c(ccc6c(cccc56)OCN5C</chem>	0.8000	0.9062	<chem>CN1CC=CCC(c2cc3cccc3o2)C1</chem>	0.9100	0.1818	0.7580
<chem>COc1cnc(O)cccsc2cccc1-2</chem>	1.0000	0.8961	<chem>COc1ccc(-c2c(-c3ccc(O)cc3O)n[nH]c2C)</chem>	0.9483	0.3077	0.8226
<chem>O=Cc1c2ccc3c4cc(cc13)C(=C2)Oc1cccc1</chem>	0.8000	0.8933	<chem>O=c1cc(-c2ccccc2)[nH]c2cc(Cc3cccc3)</chem>	0.9422	0.1864	0.6469
<chem>CNC(=O)C=C1NCC(C)CCCNC(C)C23CCCC1N2</chem>	0.8000	0.8904	<chem>Cc1sc(CN2CCNC2)cc1C(=O)NCC12CC3CC(C</chem>	0.9669	0.2121	0.5037
<chem>CC(=O)NC=Cc1cnccl</chem>	0.8000	0.8885	<chem>CC(=O)Nc1cccc1</chem>	0.9064	0.3429	0.7093
<chem>CC(=O)NCCC12CCCN3C(c4cccc4C=O)=CC=C</chem>	0.8000	0.8810	<chem>CCC(=O)NCC[C@@H]1CCc2ccc3c(c21)CCO3</chem>	0.8840	0.2283	0.7535
<chem>CC=C(C)NC(=O)C=CC=C1COCC(C)CCN(C)C1=</chem>	0.8000	0.8725	<chem>CCCC(=O)Nc1ccc(OCC(O)CNC(C)C)c(C(C)=</chem>	0.8806	0.2045	0.8117

Table 9: Measured accepted and first-failure rejected examples (strings shortened for layout).

Type	T	Generated SELFIES	Decoded SMILES	Stage	Reason
accepted	0.8000	[C][N][C][C][C][C][Branch2][Ring1][Ring2]	C1NCCC=C(NC(=O)NC=CC2=CC=CC=C2C=C1)C=O		
accepted	0.8000	[N][C][C][C][C][C][C][C][Ring1][Branch1][	N1C2=CC=CC=C1OC(=O)NC=C2C=CC3=CCC=CC=C3		
accepted	0.8000	[C][C][Branch1][C][C][C][N][C][C][C][Bran	C1C(C)CNC=CC(C1)=C1CCCC		
accepted	0.8000	[O][C][Branch1][S][C][C][C][C][Ring1][=	O=C(C=CC1=CCC=CC=C1CC1)NCCCCOCCNC2=CCCCO2		
accepted	0.8000	[C][C][C][C][C][C][Ring1][Branch1][C][C][C]	C1C=CC=C1CCCCN=C=C2C(=O)CC=C2C=C		
accepted	0.8000	[N][C][C][C][C][C][C][C][C][C][C][C][C]	N1C(=O)C=CC=CC=CC=C(C(F)N1		
rejected	0.8000	[C][C][C][C][C][C][C][C][Ring1][Branch1][	CC1=CC=CC=C1C=O	heavy_atom_range	outside_training_1st_99th_percentiles
rejected	0.8000	[C][C][C][C][C][C][C][C][Ring1][Branch1][	CC1=CC2=CC=C1C=C2C=O	molecular_weight_range	outside_training_1st_99th_percentiles
rejected	1.0000	[O][C][Branch1][C][C@@H1][Branch1][C][N][	O=C([C@@H1])CNCC(N)(CC)C(F)C=CCF	radical_electrons	nonzero_radical_electrons



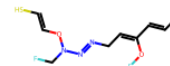
1. T=0.8; max Tan=0.08



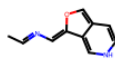
2. T=0.8; max Tan=0.09



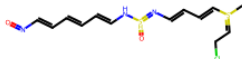
3. T=1.0; max Tan=0.10



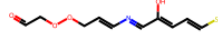
4. T=1.2; max Tan=0.10



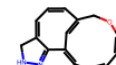
5. T=1.2; max Tan=0.10



6. T=1.2; max Tan=0.10



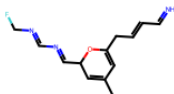
7. T=1.2; max Tan=0.10



8. T=0.8; max Tan=0.11



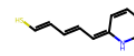
9. T=0.8; max Tan=0.11



10. T=1.2; max Tan=0.11



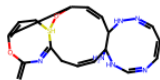
11. T=1.0; max Tan=0.11



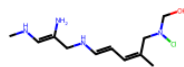
12. T=1.0; max Tan=0.11



13. T=1.2; max Tan=0.11



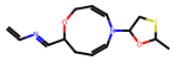
14. T=1.2; max Tan=0.11



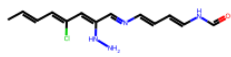
15. T=0.8; max Tan=0.11



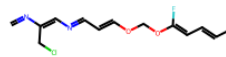
16. T=1.2; max Tan=0.11



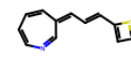
17. T=1.2; max Tan=0.11



18. T=0.8; max Tan=0.12



19. T=0.8; max Tan=0.12



20. T=1.2; max Tan=0.12

Figure 7: Up to 20 unique accepted exact-novel molecules, preferring structurally novel examples. This execution displayed 20 molecule(s).

12 Discussion

Pooling. Global mean pooling is required to map variable atom counts to one fixed-width molecular vector, which makes pairwise retrieval and the $B \times B$ contrastive matrix well-defined.

Causal learning and generation. The causal mask blocks all future tokens during next-token prediction. Teacher forcing learns conditionals from observed prefixes; iterative sampling composes those conditionals without a ground-truth continuation. Warm-up reduces the difficulty of introducing multimodal alignment before basic SELFIES patterns are learned.

Batch negatives. Batch size 256 offers roughly 255 in-batch negatives per full-batch item. Larger batches improve negative diversity but consume more graph, Transformer, and similarity-matrix memory.

Representation compatibility. Test top-1 retrieval was 0.8190 graph-to-sequence and 0.7905 sequence-to-graph, compared with random 0.0024. Matching cosine averaged 0.8922, versus 0.0129 for nonmatches. Both top-1 directions exceeded the finite-test-set random rate; therefore compatibility is not overstated. Generated graph-sequence cosines in Table 8 offer a separate out-of-training-distribution check but do not substitute for retrieval.

13 Limitations and Responsible Interpretation

A SELFIES string that decodes and passes RDKit parsing is only formally machine-readable under the implemented rules. It does not demonstrate physical or chemical stability, synthesizability, safe handling, low toxicity, biological activity, therapeutic effectiveness, environmental acceptability, legal suitability, or medical usefulness. The percentile filters inherit the training distribution and are not experimental chemistry. Fingerprint novelty is representation-dependent, and QED is a heuristic descriptor rather than a validation of usefulness. The scalar bond approximation omits rich edge chemistry, random splitting does not test scaffold extrapolation, and Lipophilicity is small relative to modern molecular pretraining corpora. This is the assignment-compliant execution with 15 language-model warm-up epochs and 300 generation attempts per temperature; joint training used validation monitoring and executed 40 epoch(s) (cap 40, patience 7).

14 Runtime

Table 10: Measured phase runtimes. Installation and dataset download/load are excluded from the total.

Phase	Seconds	Included in total
dataset_download_and_load	0.0855	False
dataset_preparation	8.1003	True
language_model_warmup	522.8845	True
joint_training	1565.5097	True
retrieval_evaluation	0.5917	True
generation	12.8415	True
validation	0.3727	True
total_excluding_installation_and_download	2110.3005	True

The experiment total excluding installation and dataset download/load was 2110.300 seconds on device `cpu`. Phase timings include dataset preparation (8.100 s), language-model warm-up (522.885 s), joint training (1565.510 s), retrieval (0.592 s), generation (12.842 s), and validation (0.373 s).

15 Conclusion

The implementation provides an auditable paired molecular pipeline with exact graph–sequence alignment, assignment-specified architectures and objectives, checkpoint-selected retrieval, autonomous SELFIES generation, strict staged validation, canonical novelty, fingerprint comparison, and generated-molecule representation analysis. Conclusions are deliberately limited to the measured run mode and do not make claims of chemical or medical utility.